

AN1.2 - AT24 Agent Framework: Locked Decisions & Architecture Decision Record

Sprint: AT24 AI Agents - Agent Framework Foundation **Stage:** Post-A0 decision lock (precedes A1) **Inputs:** AN1.1-agent-framework-architecture-audit.md + product owner review (2026-09-05) **Status:** 4 decisions LOCKED - 3 questions PENDING owner sign-off - A1 not started

This document freezes the architectural decisions coming out of the AN1.1 audit so implementation (A1->A4) cannot drift. Any change to a LOCKED decision requires an explicit new ADR entry here, not an inline code choice.

0. Core principle (LOCKED - governs everything below)

The AT24 Agent Framework is an orchestration + persistence + governance layer. It does NOT rebuild the existing Intelligence Pipeline or Quant Engine - it drives them through an AT24-owned Tool Registry.

Concretely:

- `RealTimeIntelligenceService`, `MarketIntelligencePipelineService` (v15D.12.0), the regime/hypothesis engines, `at24-quant-engine`, `services/algo-test`, `news.service`, `KnowledgeRetriever`, the risk engine and paper-trading are **called as tools**. Their source is not modified by this sprint.
- No second backtest engine, no second market-intelligence pipeline, no second evidence model. Agent evidence reuses the pipeline's `EvidenceItem` vocabulary.
- The LLM (`lib/ai AIService`, vendor-neutral) is used **only** for planner assistance and final narrative synthesis, always behind an integrity check. Never for facts, tool results, or evidence.

Rationale: this is the differentiator. AT24's value is verifiable, evidence-backed intelligence built over 15 sprints. An agent layer that re-implements any of it would be strictly worse and would fork the truth.

1. LOCKED DECISIONS

D1 - AgentType: validated free-string, registry-driven (not a DB enum)

- `Agent.type` stays a `String` column (as today). **No Prisma enum, no migration of the 3 seeded rows.**
- A code-level `AGENT_TYPE_REGISTRY` (`types/agent-framework/agent-type-registry.ts`) is the authority: it defines each type's default tools, default permission grants, and autonomy cap.
- Writes `validate type` against the registry keys. Unknown type -> 422.
- Adding `EXECUTION`, `COMPLIANCE`, `PORTFOLIO_OPTIMIZER`, `STRATEGY_MONITOR`, `DATA_QUALITY` later = one registry entry + optional tools. **Zero schema migration.**
- Legacy seeded values (`market-analyst`, `risk-manager`, `strategy-generator`) are mapped to the new registry keys by a lookup table in the registry module - the rows are not rewritten.

D2 - Tests: existing `scripts/validate-*.ts` house style (no Vitest)

- No new test-runner dependency this sprint.
- Every framework test ships as `scripts/validate-agent-*.ts` - a standalone `node:assert/strict` harness run via `tsx`, following `scripts/validate-decision-context.ts` exactly (build deterministic fake inputs -> run the real unmodified service chain -> assert -> print pass/fail).
- Each new script gets a `validate:agent-*` entry in `package.json`.
- Required coverage (from brief sec 24): contract valid/invalid, runtime success/failed/timeout/step-limit/tool-limit/credit-limit, permission allowed/denied, live-execution-denied-by-default, tool

valid/invalid/error/ timeout, evidence captured/provenance-retained/missing-handled, memory scoped read/write + policy enforcement, credit cost/usage/limit, evaluation attached/failed.

D3 - Mock agents are a legacy compatibility layer; new code is isolated

- `services/agents/*`, `types/agent.ts`, `data/mock-agents.ts`, `config/agent.config.ts`, `services/api/AgentsApi.ts`, `components/agents/*`, `app/dashboard/agents/page.tsx` - **treated as a frozen legacy layer**. Not deleted, not extended, kept working.
- All new implementation lives under:
- `types/agent-framework/` - contracts
- `services/agent-framework/` - runtime, supervisor, planner, tools, evidence, memory, credits, guardrails, evaluation, observability
- `app/api/private/agents/**` - new route handlers (the existing GET route is preserved; new endpoints are added alongside it)
- The existing `/dashboard/agents` UI is **not touched until A15**, and only after G07.
- The new AgentRun-centric models coexist with the legacy AgentTask/AgentActivity tables - legacy tables are read by the old UI, new tables by the new runtime. No cross-writes.

D4 - `research.web_search` is in v1 scope, but ONLY as a Tool Registry adapter

- The Research Agent is **not** limited to knowledge-base + news. Web search is a first-class research tool.
- **It is not hardcoded into the Agent Runtime**. Path is strictly:

Research Agent -> Planner / Supervisor -> Tool Registry (`research.web_search`) -> pluggable SearchProvider adapter -> AgentEvidence (citations + provenance mandatory)

- `research.web_search` is a ToolDefinition like any other: `inputSchema` / `outputSchema` / `requiredPermissions`: `[CAN_RUN_RESEARCH]` / `creditCost` / `handler`.
- The underlying provider/engine is replaceable behind a SearchProvider interface (same DIP pattern as `lib/ai/AIProvider` and `MarketDataProvider`). No vendor named in the contract.
- Every result item that informs a conclusion **must** produce an AgentEvidence row with source, sourceId (URL), timestamp, provenance (provider, retrievedAt), and relevance/confidence. A conclusion citing web content with no evidence row fails the output integrity check.
- Outbound network egress from `research.web_search` is subject to the same privacy rules as the rest of the platform (no user PII in query strings, no personal-data compilation).

2. PENDING - owner sign-off required before the affected step

These are the AN1.1 sec 13 questions not yet locked. Recommended defaults are given; each is low-risk, but the owner asked to review sec 13 line-by-line and return a final decision sheet.

P1 - Migration application timing (*affects A3*)

Recommendation: generate the migration + hand-reviewed SQL in A3, but **do not apply** to any live DB until an explicit go-ahead (M8/M11 precedent). Apply via `prisma migrate deploy` / reviewed SQL - **never `prisma migrate dev`** (pgvector-reset trap, see `project_sprint15c_workflow`). All new columns on Agent/AgentMemory are nullable + defaulted, so no backfill.

P2 - Run execution model (*affects A4*)

Recommendation: `POST /agents/:id/runs` executes **synchronously** for fast agents (Research / Market Intelligence on cached data - one serverless request) and returns 202 Accepted + a runId for slow agents (backtest sweeps), with continuation driven by the existing Vercel cron seam + `AgentRuntime.tick(runId)` (resumable state machine). **No dedicated queue/ worker service this sprint** (brief sec 26 - no microservices). A first-class job queue is a later, separately-scoped decision.

P3 - Credit numbers (*affects A9*)

Recommendation: no pricing decisions this sprint. Balance ceiling = `PLAN_LIMITS[planId].aiCredits` (already exists). Per-tool `creditCost` values are placeholder constants in one file (`services/agent-framework/credits/tool-credit-costs.ts`), each flagged `// PLACEHOLDER - pricing pass required`. The **mechanism** (pre-flight estimate, per-step debit, immutable ledger, hard stop) is fully built and tested; only the numbers are deferred.

3. Locked execution sequence

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AN1.1 Audit (done, committed)
|
AN1.2 Lock decisions (this doc)
|
[owner returns final decision sheet on P1-P3]
|
A1 - Locked-decision capture + Agent Contract
  - this ADR is the decision capture; A1 code = types/agent-framework/
    contracts (definition, run, step, tool-call, evidence, memory,
    permission, credit, evaluation) + contractVersion "AF-v1"
    + agent-type-registry.ts + validate-agent-contracts.ts
|
A2 - Tool Contract + Tool Registry
  - ToolDefinition, ToolRegistry (code Map), ToolGateway
    (schema-validate -> permission-check -> autonomy-check ->
    credit-check -> invoke handler)
  - v1 tool stubs wrapping existing engines (see AN1.1 sec 3.3)
  - validate-agent-tools.ts
|
A3 - Agent Run / State Contract + Prisma migration (generated, NOT applied)
  - 7 new models + additive Agent/AgentMemory columns
  - reviewed SQL ready; application gated on P1 go-ahead
|
A4 - Server-side resumable tick() Runtime
  - AgentRuntime: load -> policy check -> supervisor/plan ->
    tool loop -> evidence capture -> state persist -> output
  - LimitEnforcer (max steps / tool calls / runtime / retries / credit)
  - RunTracer (structured trace -> AgentStep rows + logger)
  - validate-agent-runtime.ts (success + every failure/limit path)
|
[Gate G02 review] then A5+ per AN1.1 sec 7
```

Not in this window: any change to `/dashboard/agents`, any real agent (A11-A13), any UI work, any live-execution wiring.

4. Invariants carried into implementation (LOCKED)

- Runtime is **server-only** (`services/agent-framework/*`) - never in the client bundle. Client gets a typed API client only.
- `AgentStep` / `AgentToolCall` / `AgentEvidence` / `AgentCreditLedgerEntry` are **append-only, immutable, no deletedAt, no update path** (mirror `AuditLog` / `AgentActivity`). A re-run is a new `AgentRun` row.
- `PermissionPolicy` is an **allowlist; default deny**. `CAN_CREATE_ORDER` / `CAN_EXECUTE_ORDER` are rejected at the policy-parser layer unless a server env flag **and** a user allowlist both pass - `LIVE_EXECUTION = DENY` enforced twice (parser + tool gate).
- Default `autonomyLevel = 0`. Trading Decision Agent capped at L1 (decision-support) in v1. L3/L4 are contract-only - no execution wiring this sprint. L2 = existing paper-trading service.
- Every tool invocation passes `CreditGateway` - **no bypass path**. A run that would exceed `min(remaining, perRunCeiling, perDayCeiling)` ends `credit_limit` with partial output + full trace preserved.
- Final agent output is validated against `AgentDefinition.outputSchema` **and** an evidence-integrity check (reuse the `ai-response-integrity.service.ts` pattern) - a claim with no backing `AgentEvidence` row is rejected.

- 7. No vendor name in any contract type. modelPolicy is { preferred, fallback[], allowed[] }. LLM access only through lib/ai AIService.
- 8. AT24 owns tool **selection** (planner/supervisor) - provider-native tool-calling is not relied upon (neither provider layer supports it today; can be added later without a contract change).

5. Change log

Date	Entry
2026-09-05	AN1.2 created. D1-D4 locked from AN1.1 + owner review. P1-P3 pending owner decision sheet. Core principle (sec 0) and invariants (sec 4) locked.